Recoil transport and environmental exposure of black-hole merger remnants

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Black-hole mergers in stellar clusters can leave remnants moving through their hosts after receiving gravitational-wave recoil. Remaining gravitationally bound does not determine how soon a remnant enters a dense region or how much time it spends there. Predicting this motion requires a stellar environment that consistently supplies gravity, collisional drift, and velocity diffusion. We develop a merger-conditioned transport formulation and define finite-region residence and species-resolved environmental exposure on its trajectories. We then examine its observational inputs using public astrometric and structural products and published multimass-model posteriors for six Galactic clusters. Native measurements are compared in common physical and dimensionless units while preserving tracer selection and joint parameter dependence and recording shared distance uncertainty. The analysis identifies restricted common radial sampling, strong mass–radius correlations in several host posteriors, and differences between published structural estimates that cannot be interpreted as cluster evolution. These findings establish a source-aware basis for constructing candidate environments and identify the local stellar distributions still needed to evaluate transport. The reported numerical results concern observations and existing host inferences; remnant trajectories, friction coefficients, and residence distributions remain to be calculated.

I. INTRODUCTION

A black hole formed by a merger can receive a recoil from anisotropic gravitational-wave emission. Within a stellar cluster, its subsequent motion reflects both that recoil and the state of the system in which the merger occurs. The progenitor's center-of-mass motion, merger position, host potential, and gravitational scattering determine the remnant's evolving position and velocity. A gravitationally bound remnant need not remain near the center, and entry into a dense region does not imply that the remnant comes to rest there.

The distinction matters whenever the surrounding population influences subsequent evolution. Residence measures the time spent in a specified region, whereas density exposure weights the trajectory by the material encountered along it. Neither is fixed by a retention flag alone. Their calculation requires a dynamical description with enough information to distinguish repeated excursions, continued wandering, and changes in the environment during the remnant's motion.

The necessary kinetic ingredients are established. Rosenbluth collision integrals relate velocity drift and diffusion to a field-particle distribution [1]; their gravitational application permits anisotropic velocity distributions [2]. Direct calculations of kicked black holes illustrate limitations of simplified dynamical-friction descriptions and show that individual encounters can increase a remnant's energy [3]. These results motivate treating mean drift and fluctuations together, while testing the regime in which a local diffusion closure is adequate. They do not supply a calibrated prescription for an arbitrary cluster.

We formulate two linked problems. Step 1 defines merger-conditioned transport in position and velocity, with an explicit matter inventory, reference frame, collision closure, and termination rule. Step 2 defines finite-region entry, residence, and species-resolved environmental exposure on the resulting histories. Their common output is a timed description of the remnant's occupation of its environment. The immediate question is which launch and host information must be retained to predict these observables at a declared accuracy.

We connect this formulation to an analysis of six observed Galactic clusters. Proper-motion profiles, structural estimates, and published multimass-model posteriors permit concrete comparisons of host scales and parameter dependence. Distribution-function families can supply candidate consistent hosts [4], but measured projected kinematics are not themselves local phase-space distributions. Our numerical findings concern this observational connection: which quantities can be compared, which dependencies must be preserved, and which inputs remain unestablished. We do not refit the clusters or infer recoil, friction, or residence outcomes from these comparisons.

II. OBSERVATIONAL INPUTS AND COMPARISON METHOD

A. Sample and source distinctions

We examine NGC 0104 (47 Tuc), NGC 1851, NGC 3201, NGC 5139 (ω Centauri), NGC 6121 (M4), and NGC 6397. This deliberately selected sample provides varied mass and size scales and public kinematic and model products. It is not a complete cluster census or a selection-corrected population of black-hole-merger hosts.

For five clusters we use the original binned Hubble Space Telescope (HST) proper-motion products of Libralato et al. [5]. For ω Centauri we use the logarithmic radial bins released with Häberle et al. [6]. The former dispersion profiles use bright subgiant and red-giant tracers, whereas the latter use the full magnitude range. Stellar mass and selection can affect dispersion in a partially relaxed system; converting the units cannot remove this distinction.

We inspect the original Gaia EDR3 field catalogs and fitted radial curves of Vasiliev and Baumgardt [7], separately

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from compiled kinematic profiles. Membership probabilities, quality flags, measurement errors, and source identifiers describe the field catalogs. They do not make every listed star a cluster member, and raw proper-motion spread is not an intrinsic velocity dispersion. The released smooth radial curves are correlated fit products; their grid points are not independent observational bins.

Structural comparison values are taken from one captured path of the Baumgardt cluster database associated with the modeling and distance studies in Refs. [8, 9].[1] These values are model-derived estimates. In particular, a three-dimensional half-mass radius is distinct from a projected half-light or core radius. The compiled surface profiles mix normalized stellar counts and, for some sources, surface brightness; they are not directly a three-dimensional mass density.

For a separate view of host uncertainty, we inspect the released standard-fit multimass posterior samples discussed by Dickson et al. [10, 11]. Original observations, compilations, and these posterior products share underlying evidence. We keep the views separate rather than multiply them together as independent likelihood factors. The complete data collection comprises 51 captured files, including supporting source definitions and historical candidate observation bundles.

B. Native measurements, units, and shared uncertainty

For angular radius α, distance D, and proper-motion dispersion σ_μ, the small-angle conversions are

$$\frac{R}{\mathrm{pc}} = \frac{1000}{206264.806}\,\frac{D}{\mathrm{kpc}}\,\frac{\alpha}{\mathrm{arcsec}}, \tag{1}$$

$$\frac{\sigma_\perp}{\mathrm{km\,s^{-1}}} = 4.74047046\,\frac{D}{\mathrm{kpc}}\,\frac{\sigma_\mu}{\mathrm{mas\,yr^{-1}}}. \tag{2}$$

The radial and tangential proper-motion components are projected components, not radial and tangential velocities in three-dimensional space. Line-of-sight dispersions already expressed in physical velocity units require no proper-motion conversion. Native measurements and asymmetric errors are retained alongside the display conversions.

For a vector of converted quantities $\mathbf{y}(D)$, a shared distance error contributes, to first order,

$$C_{ab}^{(D)} = \frac{\partial y_a}{\partial D}\,\frac{\partial y_b}{\partial D}\,\sigma_D^2. \tag{3}$$

This is one covariance contribution, not the complete uncertainty model. Independence of distance and the measured quantities is not guaranteed, particularly for a kinematic distance. Joint propagation must retain any cross-covariances. Our plots condition on a catalog distance and show converted measurement errors only; distance derivatives and uncertainty are recorded separately. They are not added as independent errors in each radial bin.

Using total mass M and three-dimensional half-mass radius r_h from the same catalog view, define comparison scales

$$r_0 = r_h, \qquad v_0 = \sqrt{\frac{GM}{r_h}}, \qquad t_0 = \frac{r_h}{v_0}. \tag{4}$$

We use $G = 4.30091727 \times 10^{-3}\,\mathrm{pc\,(km\,s^{-1})^2}\,M_\odot^{-1}$ and convert $1\,\mathrm{pc}/(\mathrm{km\,s^{-1}}) = 0.97779222\,\mathrm{Myr}$. The quantities R/r_h and $\sigma_\perp/v_0$ provide common comparison coordinates. This does not impose a shared density profile or make the clusters homologous. Neither v_0 nor t_0 is a measured dispersion, escape speed, or relaxation time. Uncertainty in the joint mass, radius, and distance scales is not propagated into these display plots.

C. Joint host parameters

The inspected posterior products share the coordinate pattern

$$\begin{aligned}\boldsymbol{\theta}_c = (&W_0, M, r_h, \ell_a, g_\mathrm{t}, \delta, s^2, F,\\ &\alpha_1, \alpha_2, \alpha_3, f_\mathrm{ret}, D)_c,\end{aligned} \tag{5}$$

for cluster c. Here ℓ_a is the base-10 logarithm of the dimensionless anisotropy radius, g_t controls truncation, and δ governs the mass dependence of the velocity scale. These quantities belong to the specified lowered-isothermal and stellar-evolution model [4, 10]. The mass-function slopes α_i and black-hole retention parameter f_ret retain that model's conventions. The stored retention parameter is a percentage, not the present-day black-hole mass fraction. The coordinates s^2 and F are likelihood nuisance parameters rather than forces or velocity factors. The symbol g_t is separate from the geometric region function and Rosenbluth potentials below.

We normalize each stored log weight ℓ_k^w as

$$w_k = \frac{\exp(\ell_k^w - \ell_\mathrm{max}^w)}{\sum_l \exp(\ell_l^w - \ell_\mathrm{max}^w)}. \tag{6}$$

Moments and marginal quantiles are computed directly from complete weighted rows, without resampling or evaluating a host model. For example,

$$\overline{\boldsymbol{\theta}} = \sum_k w_k \boldsymbol{\theta}_k, \tag{7}$$

$$\mathsf{C}_\theta = \sum_k w_k (\boldsymbol{\theta}_k - \overline{\boldsymbol{\theta}})(\boldsymbol{\theta}_k - \overline{\boldsymbol{\theta}})^\mathsf{T}. \tag{8}$$

This is the covariance of the weighted empirical posterior; it is not a finite-sample unbiased estimator. Marginal quantiles use interpolation in the sorted weighted cumulative distribution. Nuisance coordinates remain in the joint records, and marginal summaries do not define independent parameter draws.

[1] The selected structural table is the database's /newdata/ parameter table. The root-path table is retained as a separate comparison; values from the two paths are not merged.

TABLE I. Source coverage. Gaia field rows include nonmembers. HST bins refer to the original proper-motion profiles; the combined, radial and tangential quantities share stars. Flagged rows belong to the separate 418-row compilation and are not corrected or used as likelihood inputs.

Cluster	Gaia field rows	HST bins	Flagged rows
47 Tuc	144,761	28	0
NGC 1851	12,688	20	4
NGC 3201	63,171	6	0
ω Cen	228,055	40	1
M4	69,417	6	0
NGC 6397	103,786	7	15
Total	621,878	107	20

Exact observation-file and software-revision identifiers were not found in the inspected run metadata. Historical candidate inputs are therefore documented without claiming a reproduced fit. Current resources cannot automatically replace the inputs to an older posterior. The distinct NGC 6397 no-black-hole alternative discussed in Ref. [11] has a released summary and figures but no corresponding chain in the inspected release.

III. OBSERVATIONAL FINDINGS

A. Coverage and source reconciliation

Table I summarizes the inspected products. The six original Gaia field catalogs contain 621,878 rows, with no repeated source identifier within a cluster. The HST products provide 107 radial bins and 321 component records. Combined, projected radial, and projected tangential dispersions in a bin share stars; the combined value is not a third independent measurement. The released Gaia curves contain 101 grid points per cluster, or 606 correlated samples in total.

The separate compilation contains 418 kinematic rows, of which 20 require reconciliation: four zero-radius entries for NGC 1851, one all-zero entry for ω Centauri, and 15 NGC 6397 entries whose dispersion equals both quoted errors. Four NGC 1851 records also share one radius, source, and component key. These observations motivate flags, not guessed corrections or a claim that every flagged value is physically impossible. The compilation is retained for comparison and is not treated as an approved likelihood input. The 422 surface-profile records likewise retain their unresolved absolute mass normalization.

Figure 1 shows the variation in radial sampling. At the catalog point scales, the intersection of the six HST median-radius spans is approximately 0.081–0.251 in projected R/r_h. This is not exact shared annular coverage: most HST bin edges are unavailable, and a span does not establish sampling density between its endpoints. No common radial interpolation or likelihood is constructed.

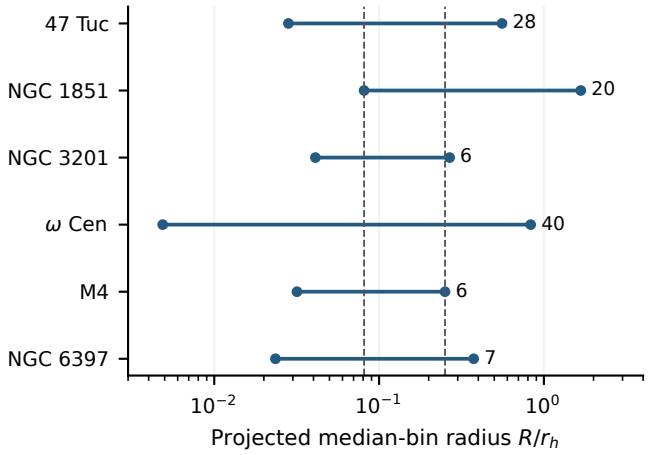


FIG. 1. Spans of original HST median-bin radii in common comparison coordinates. Numbers at the right give radial-bin counts. Dashed lines mark the intersection of the six spans at fixed catalog scales. They do not represent bin edges or a common three-dimensional domain.

B. Structural scales and projected kinematics

Table II reports the selected structural estimates and derived scales. Catalog mass ranges from $8.88 \times 10^4\,M_\odot$ for M4 to $3.32 \times 10^6\,M_\odot$ for ω Centauri; the listed central escape speeds range from 18.6 to 61.9 km s^{-1}. These are conditional model estimates, not a measured retention law for merger remnants. Retention would additionally depend on position, inherited progenitor motion, recoil direction, external gravity, and the criterion used to classify escape.

The derived time scales span approximately 0.155–0.851 Myr. They establish dimensional comparison units, not return or damping times. Radius and escape-speed errors are not supplied with these catalog entries; the quoted mass and distance errors also do not capture every model systematic. Mixing a catalog mass with a radius or distance from a different fit would create an unsupported joint parameter combination.

Figure 2 compares original HST radial and tangential dispersions at the catalog scales. It displays the available projected constraints without fitting a common profile. Tracer selection, scale uncertainty, calibration, and radial sampling differ across clusters. Consequently the figure neither demonstrates universal kinematics nor statistically rejects them. HST transformations can remove sky-plane streaming signals [5]; absence of such a signal in these dispersion products is not evidence of zero cluster rotation.

C. Posterior dependence and model sensitivity

Table III reports selected marginal summaries of the published standard-fit samples. The effective sample sizes, calculated as $1/\sum_k w_k^2$, reproduce all six printed source values and range from approximately 5,171 to 6,889. Twelve param-

TABLE II. Structural-catalog estimates and derived display scales. The selected catalog path supplies the quoted mass and distance errors, but no errors for r_h or $v_{\mathrm{esc},0}$; missing uncertainty is not zero. Model systematics and joint scale uncertainty are not included. The gravitational scales v_0 and t_0 are defined in Eq. (4); t_0 is not a friction time.

Cluster	D (kpc)	$M/(10^5 M_\odot)$	r_h (pc)	$v_{\mathrm{esc},0}$ (km s^{-1})	v_0 (km s^{-1})	t_0 (Myr)
47 Tuc	4.52 ± 0.03	8.900 ± 0.050	6.24	51.0	24.77	0.246
NGC 1851	11.95 ± 0.13	2.780 ± 0.040	3.11	42.7	19.61	0.155
NGC 3201	4.74 ± 0.04	1.800 ± 0.030	8.37	18.6	9.62	0.851
ω Cen	5.43 ± 0.05	33.200 ± 0.100	9.85	61.9	38.07	0.253
M4	1.85 ± 0.02	0.888 ± 0.011	3.81	19.5	10.01	0.372
NGC 6397	2.48 ± 0.02	0.901 ± 0.009	3.17	23.4	11.06	0.280

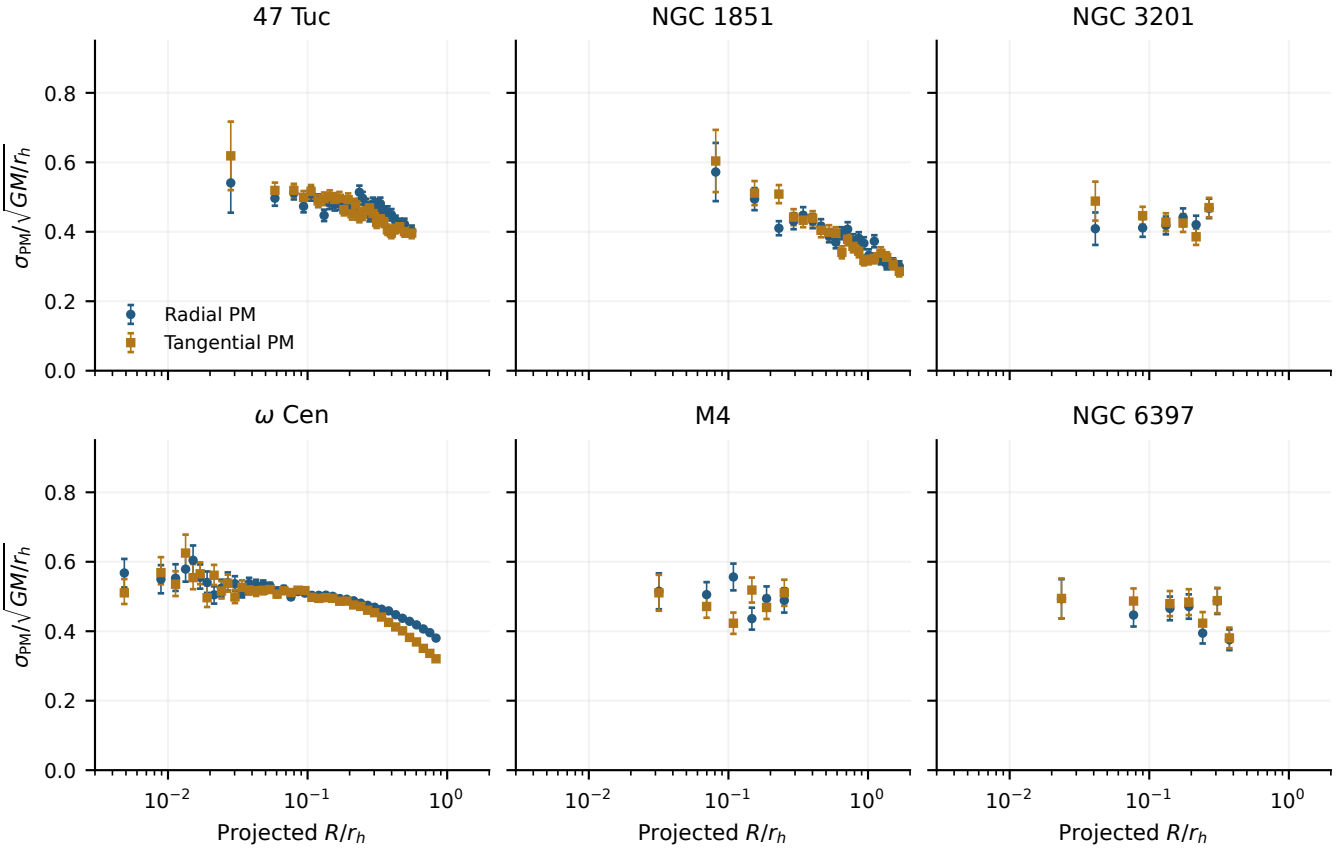


FIG. 2. Original projected radial and tangential HST proper-motion dispersions in common dimensionless coordinates. Circles and squares distinguish components. Five clusters use the bright-tracer profiles of Libralato et al. [5]; ω Centauri uses the full-magnitude-range profiles of Häberle et al. [6]. Error bars retain the source measurement-error convention and condition on catalog point scales. Shared distance and joint structural uncertainty are not fully propagated. Components and bins need not be statistically independent.

eter rows differ beyond three-decimal rounding in a mean or standard deviation. The original summary procedure averages statistics from resampled runs, whereas Eqs. (6)–(8) describe the original weighted sample. This estimator distinction permits close agreement without rounding-identical moments; it does not establish exact source-fit reproduction.

Mass and radius have weighted correlations of 0.900 in NGC 3201 and 0.944 in ω Centauri. Independent sampling of their marginal distributions would discard these dependencies. Because the scales in Eq. (4) involve ratios and products of these parameters, uncertainty should be propagated from complete host realizations. This requirement also applies to quantities derived from a realization, including its potential and species distributions.

Figure 3 compares the published multimass marginal estimates with the structural catalog. For example, NGC 6397 has catalog $r_h = 3.17$ pc, compared with a standard-fit posterior median 4.887 pc and marginal 16th–84th interval [4.747, 5.044] pc. The latter is not the unavailable alternative no-black-hole chain. This difference identifies sensitivity to

TABLE III. Direct weighted marginal medians and 16th–84th percentiles from the published standard-fit joint samples of Dickson et al. [10, 11]. These are conditional model estimates, not new fits. The NGC 6397 row is the standard fit; its distinct no-black-hole alternative lacks a released chain in the inspected product. Correlations refer to the complete weighted samples.

Cluster	$M/(10^5 M_\odot)$	r_h (pc)	D (kpc)	Corr(M, r_h)
47 Tuc	9.045 [8.996, 9.094]	6.714 [6.679, 6.749]	4.420 [4.408, 4.431]	0.327
NGC 1851	3.250 [3.211, 3.288]	3.712 [3.670, 3.757]	12.183 [12.106, 12.268]	-0.235
NGC 3201	1.757 [1.699, 1.814]	9.598 [9.070, 10.152]	4.636 [4.603, 4.671]	0.900
ω Cen	30.824 [30.693, 31.489]	9.419 [9.402, 9.529]	5.300 [5.287, 5.327]	0.944
M4	0.911 [0.894, 0.929]	3.920 [3.859, 3.982]	1.852 [1.838, 1.866]	0.316
NGC 6397	1.084 [1.061, 1.109]	4.887 [4.747, 5.044]	2.436 [2.422, 2.452]	0.782

model and data choices; it is neither temporal expansion measured between releases nor a statistical rejection using independent datasets.

D. What the comparison establishes

The analysis supports a common field and unit convention with a distinct joint host distribution for each cluster. It does not establish six interchangeable scattering backgrounds. NGC 3201 and 47 Tuc provide initial candidates for reproducing a published host and checking its projected observables, with rotation explicitly assessed for 47 Tuc. NGC 1851 and M4 extend the structural comparison. NGC 6397 tests sensitivity in a core-collapsed system, while ω Centauri requires particular attention to rotation, flattening, and mass-dependent tracer selection [6, 11].

Published density and cumulative-mass bands do not by themselves provide mutually consistent local host realizations. Independently assembling median profiles could violate the consistency between mass density, gravity, and velocity structure. These observational findings therefore constrain how the following transport problem should be supplied with inputs; they do not evaluate its drift, diffusion, or trajectory observables.

IV. STEP 1: RECOIL TRANSPORT

A. Matter distributions and reference frame

Use physical units and cluster-relative coordinates in translating, nonrotating axes. Let $\mathbf{R}(t)$ be the reference position in an inertial frame, and let $(\mathbf{r}, \mathbf{v})$ denote the remnant's relative position and velocity. The host description Θ contains its matter distributions, gravitational boundary conditions, and any supplied temporal evolution.

For species j with individual mass m_j, define a number phase-space distribution $F_j(\mathbf{r}, \mathbf{u}, t \mid \Theta)$. Thus $F_j\, d^3r\, d^3u$ is an expected object count. Its dimensions are inverse volume times inverse velocity cubed. The number and mass densities are

$$n_j(\mathbf{r}, t) = \int F_j(\mathbf{r}, \mathbf{u}, t)\, d^3u, \tag{9}$$

$$\rho_j(\mathbf{r}, t) = m_j n_j(\mathbf{r}, t), \qquad \rho_c = \sum_j \rho_j. \tag{10}$$

The cluster potential satisfies

$$\nabla^2 \Phi_c = 4\pi G \rho_c, \tag{11}$$

together with specified boundary conditions. The tracked remnant's self-field is excluded. An unresolved binary can be represented by its center of mass only on encounter scales supporting that approximation; its mass must not also enter through separately counted components. A continuous mass spectrum requires a mass integral or a declared binning convention.

The same distributions determine streaming and velocity dispersion:

$$\overline{\mathbf{u}}_j = \frac{1}{n_j} \int \mathbf{u} F_j\, d^3u, \tag{12}$$

$$S_{j,ik} = \frac{1}{n_j} \int (u_i - \overline{u}_{j,i})(u_k - \overline{u}_{j,k}) F_j\, d^3u. \tag{13}$$

These moments are defined where $n_j > 0$. Distinct species may have different concentrations, streaming motions, and anisotropies. Density and a scalar dispersion do not generally determine the full velocity distribution needed for a collision calculation.

For an external gravitational potential $\Phi_{\rm ext}$, the relative external acceleration is

$$\mathbf{a}_{\rm ext,rel} = -\nabla \Phi_{\rm ext}(\mathbf{R} + \mathbf{r}, t) - \ddot{\mathbf{R}}(t). \tag{14}$$

If the reference orbit follows $\ddot{\mathbf{R}} = -\nabla \Phi_{\rm ext}(\mathbf{R}, t)$, Eq. (14) is the differential tidal acceleration. There are no Coriolis terms in these nonrotating axes. Internal stellar rotation remains possible through F_j.

This is a test-remnant description in a supplied background. Poisson's equation does not evolve the host. A frozen host needs a validity interval; an evolving host needs mutually consistent distributions and potential at each epoch. Backreaction, collective response, and energy feedback require additional equations and are not obtained by introducing a time argument alone.

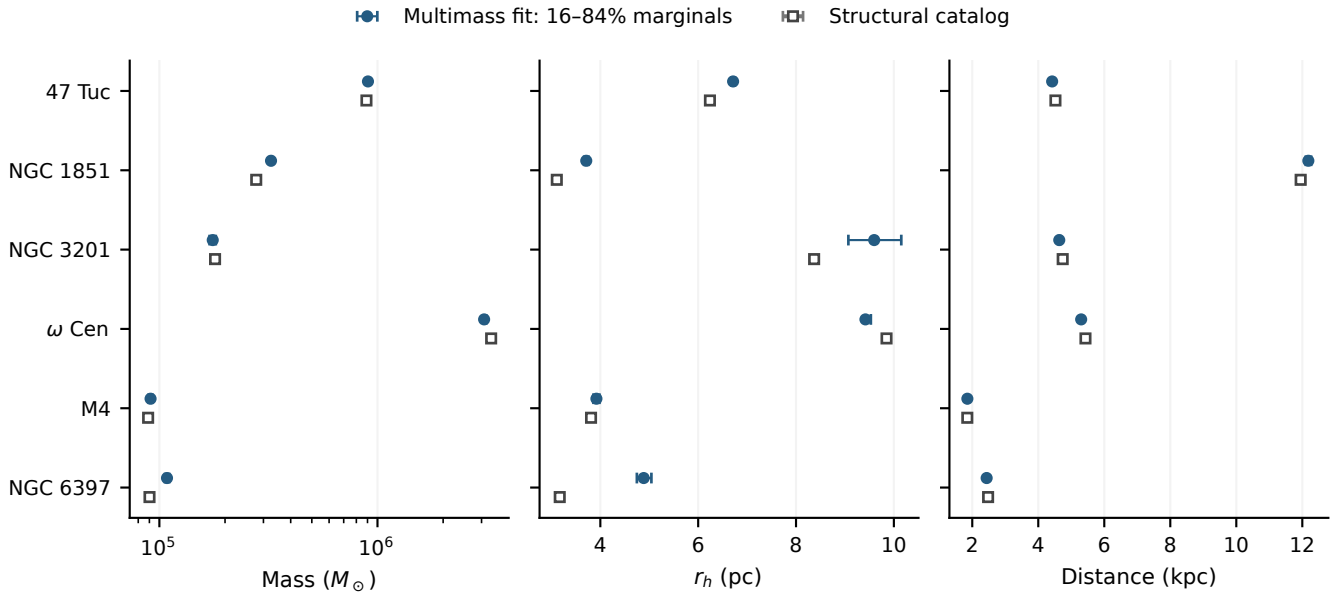


FIG. 3. Structural-catalog estimates (open squares) and published standard-fit posterior marginals (circles with 16th–84th intervals). The mass axis is logarithmic. Catalog mass and distance errors are shown where provided; catalog radius errors are unavailable. Source observations overlap, and model assumptions differ. The NGC 6397 standard fit is kept distinct from its no-black-hole alternative. The comparison measures dependence on the adopted inference, not cluster evolution.

B. Merger-conditioned initial conditions

Set elapsed time zero at the remnant-producing merger. The launch conditions are

$$\mathbf{r}(0) = \mathbf{r}_{\rm m}, \qquad \mathbf{v}(0) = \mathbf{v}_{\rm COM} + \mathbf{v}_{\rm k}, \tag{15}$$

$$m_{\rm r}(0) = m_{\rm rem}. \tag{16}$$

Here $\mathbf{v}_{\rm COM}$ is the progenitor binary's center-of-mass velocity, not its internal orbital velocity. The recoil vector $\mathbf{v}_{\rm k}$ must be transformed into the cluster frame before addition. Equation (15) uses nonrelativistic cluster kinematics; a relativistic merger calculation supplies recoil and remnant mass.

Let $\mathcal{D}$ denote observational or simulation constraints and $\mathcal{M}$ the merger-selection rule and relevant assembly information. The initial law is joint:

$$z_0 = (\mathbf{r}_{\rm m}, \mathbf{v}_{\rm COM}, \mathbf{v}_{\rm k}, m_{\rm rem}), \tag{17}$$

$$z_0 \sim \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}). \tag{18}$$

No independence among these variables or from the host is imposed. In particular, the initial specific angular momentum $\mathbf{r}_{\rm m}\times(\mathbf{v}_{\rm COM}+\mathbf{v}_{\rm k})$ need not vanish. A noncentral trajectory need not pass through the exact center.

A retained-only population is meaningful only after specifying the escape criterion and the interval over which retention is assessed. Initial negative cluster-relative energy does not by itself establish permanent retention in an evolving tidal environment. The base transport process is defined before any selection on future survival. Statistics conditioned on that survival require conditioning or reweighting its full path law; changing the launch distribution alone while leaving subsequent stochastic evolution unchanged is insufficient.

C. Velocity drift and diffusion

Define the smooth acceleration $\mathbf{a}_g = -\nabla\Phi_{\rm c} + \mathbf{a}_{\rm ext,rel}$. A local diffusion description takes the Itô form

$$d\mathbf{r} = \mathbf{v}\,dt, \tag{19}$$

$$d\mathbf{v} = (\mathbf{a}_g + \mathbf{A})\,dt + \mathbf{B}\,d\mathbf{W}, \tag{20}$$

$$\mathbf{D} = \mathbf{B}\mathbf{B}^{\mathsf{T}}. \tag{21}$$

The three Wiener components satisfy $\mathbb{E}[dW_i\,dW_k] = \delta_{ik}\,dt$. After subtracting smooth gravity, the first and second collisional increments define

$$A_i = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll} \mid \mathbf{r}, \mathbf{v}, t]}{\Delta t}, \tag{22}$$

$$D_{ik} = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll}\Delta v_k^{\rm coll} \mid \mathbf{r}, \mathbf{v}, t]}{\Delta t}. \tag{23}$$

These limits belong to a coarse-grained approximation; they do not resolve the duration of each gravitational encounter. The dimensions of $\mathbf{A}$ and $\mathbf{D}$ are acceleration and velocity squared per time, respectively. Once $\mathbf{A}$ is the complete Itô first moment, an additional noise-induced drift must not be appended.

Between discrete transitions, remnant mass is fixed on each trajectory. At fixed $\mathcal{Q} = (m_{\rm r}, \Theta, \mathcal{D}, \mathcal{M})$, let $p(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q})$ denote its phase-space density. The interior evolution is

$$\partial_t p = -\partial_{r_i}(v_i p) - \partial_{v_i}[(a_{g,i} + A_i)p] + \frac{1}{2}\partial_{v_i}\partial_{v_k}(D_{ik}p). \tag{24}$$

Repeated Cartesian indices are summed. The factor of one half follows from Eq. (21). Initial, boundary, and termination conditions are needed to define a solution. When launch masses vary, each mass-dependent operator must retain its own conditioning; replacing it by an operator evaluated at a mean mass is not generally equivalent.

D. A candidate weak-scattering closure

For local, uncorrelated weak deflections, a leading-logarithm Rosenbluth closure supplies both coefficients from the same matter distribution [1, 2]. At fixed position and time, define

$$h_j(\mathbf{v}) = \int \frac{F_j(\mathbf{r}, \mathbf{u}, t)}{|\mathbf{v} - \mathbf{u}|}\, d^3u, \tag{25}$$

$$g_j(\mathbf{v}) = \int F_j(\mathbf{r}, \mathbf{u}, t)|\mathbf{v} - \mathbf{u}|\, d^3u. \tag{26}$$

For the number normalization in Eq. (9),

$$A_i = 4\pi G^2 \sum_j m_j(m_\mathrm{r} + m_j) \ln\Lambda_j\, \partial_{v_i} h_j, \tag{27}$$

$$D_{ik} = 4\pi G^2 \sum_j m_j^2 \ln\Lambda_j\, \partial_{v_i}\partial_{v_k} g_j. \tag{28}$$

A mass-normalized distribution instead absorbs a factor of m_j into each Rosenbluth potential. Mixing those conventions would change the collision coefficients.

The Coulomb logarithm $\ln\Lambda_j = \ln(b_{\max,j}/b_{\min,j})$ represents a declared range of encounter impact parameters. It is held fixed during the velocity integrals and differentiations above. Arbitrary velocity-dependent cutoffs require a consistent derivation rather than an unqualified substitution. Neither cutoff values nor a prescription are selected here.

Writing $\mathbf{w} = \mathbf{v} - \mathbf{u}$, the derivative kernels are

$$\partial_{v_i} h_j = -\int F_j \frac{w_i}{w^3}\, d^3u, \tag{29}$$

$$\partial_{v_i}\partial_{v_k} g_j = \int F_j \left(\frac{\delta_{ik}}{w} - \frac{w_i w_k}{w^3}\right) d^3u. \tag{30}$$

For nonnegative distributions and positive Coulomb logarithms, $\mathbf{D}$ is symmetric and positive semidefinite. Streaming and anisotropy enter through F_j; a Maxwellian is not imposed. Mean collisional drift need not oppose the cluster-frame velocity in every anisotropic or streaming population.

The closure excludes strong encounters and collective or long-range correlations. Its applicability depends on the scales and populations contributing to relaxation. Kicked-black-hole calculations showing departures from simplified friction models [3] therefore motivate independent validation, not an assumption that adding these coefficients is sufficient.

E. Energy, termination, and probability accounting

For fixed remnant mass, define specific cluster-relative energy

$$\epsilon = \frac{1}{2}|\mathbf{v}|^2 + \Phi_\mathrm{c}(\mathbf{r}, t). \tag{31}$$

Itô's rule gives

$$d\epsilon = \left[\partial_t\Phi_\mathrm{c} + \mathbf{v}\cdot\mathbf{a}_\mathrm{ext,rel} + \mathbf{v}\cdot\mathbf{A} + \frac{1}{2}\,\mathrm{tr}\,\mathbf{D}\right] dt + \mathbf{v}^\mathsf{T}\mathbf{B}\, d\mathbf{W}. \tag{32}$$

The diffusion contribution is required. Individual trajectories can gain energy, and this energy is not a conserved tidal escape criterion. No condition resets position or velocity upon core entry.

For a common physical horizon T, define the modeled endpoint

$$T_* = \min\{T, \tau_\mathrm{esc}, \tau_\mathrm{strong}, \tau_\mathrm{valid}\}. \tag{33}$$

The stopping times represent declared escape, a strong encounter requiring an unavailable transition model, and loss of validity of the supplied description. A strong-encounter stop needs a detector, resolved objects, or a justified event model; missing encounter information is not evidence of no encounters. Weak scattering and explicitly resolved interactions must partition deflections without double-counting.

Let $p_\mathrm{alive}(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q})$ include only paths with $t < T_*$. Then

$$\int p_\mathrm{alive}\, d^3r\, d^3v = S(t \mid \mathcal{Q}) = \Pr(t < T_* \mid \mathcal{Q}). \tag{34}$$

Survival here means continued inclusion in the transport description. It does not mean that the black hole ceases to exist when a path is stopped. The transport operator requires the corresponding absorbing boundary or justified killing mechanism. A history-dependent stop requires an augmented state or a justified history-dependent treatment; it does not automatically close Eq. (24) in position and velocity.

The density in Eq. (34) is conditional on fixed remnant mass. A joint mass density is obtained by multiplication with the initial mass marginal π_m:

$$p^\mathrm{joint}_\mathrm{alive}(\mathbf{r}, \mathbf{v}, m, t \mid \Theta, \mathcal{D}, \mathcal{M}) = \pi_m(m \mid \Theta, \mathcal{D}, \mathcal{M}) p_\mathrm{alive}(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q}_m), \tag{35}$$

where $\mathcal{Q}_m = (m, \Theta, \mathcal{D}, \mathcal{M})$. Integrating a fixed-mass conditional density over mass without this weight would not represent the launch ensemble.

V. STEP 2: RESIDENCE AND ENVIRONMENTAL EXPOSURE

A. Finite-region entry and re-entry

Specify an operational region by a level-set function,

$$C(t) = \{\mathbf{r} : g(\mathbf{r}, t) < 0\}. \tag{36}$$

The function is negative inside, positive outside, and zero on the boundary. This g describes geometry and is distinct from the velocity-space potentials g_j in Eq. (26). A sphere may be represented by $g = |\mathbf{r}| - r_c(t)$, but neither a shape nor a radius is selected. A projected observational core radius is not automatically the required spatial boundary.

For an initially interior launch, define first exit and subsequent re-entry as

$$\tau_{\text{out}} = \inf\{t > 0 : g(\mathbf{r}_t, t) > 0\}, \tag{37}$$

$$\tau_{\text{re}} = \inf\{t > \tau_{\text{out}} : g(\mathbf{r}_t, t) < 0\}. \tag{38}$$

The strict opposite-side conditions exclude an isolated tangential contact. For a continuous path, the infimum identifies the boundary time associated with a sign-changing crossing. An observed re-entry delay is $\tau_{\text{re}} - \tau_{\text{out}}$.

For an initially exterior launch, define first entry by

$$\tau_{\text{ent}} = \inf\{t > 0 : g(\mathbf{r}_t, t) < 0\}. \tag{39}$$

A boundary launch requires a declared initial-side convention. Classification may use the first definite side reached, while recording the contact separately. It must not create an artificial return event. A path that never exits has no observed re-entry, and the convention $\inf \varnothing = +\infty$ does not establish that a censored event is physically impossible.

For a differentiable boundary,

$$\frac{d}{dt} g(\mathbf{r}_t, t) = \partial_t g + \mathbf{v}_t \cdot \nabla g. \tag{40}$$

Boundary evolution can change membership without inward remnant motion. Both the crossing and its origin must remain identifiable. Sign changes remain the definition even when the derivative vanishes. Separate inner and outer event surfaces can control numerical recrossings only if their effects are reported; they do not silently redefine $C(t)$.

Only events observed before the modeled endpoint are recorded as completed. Simultaneous crossing and stopping events require an ordering convention. Return flags and termination causes are separate fields: a path can re-enter the region and later escape.

B. Path-integrated observables

The residence accumulated before termination is

$$C_T = \int_0^{T_*} \mathbf{1}_{\{\mathbf{r}_t \in C(t)\}}\, dt. \tag{41}$$

It has units of time and obeys $0 \le C_T \le T_* \le T$. The ratio C_T/T uses a common horizon across realizations. If a description stops before T because subsequent motion is unresolved, this ratio is not a complete-horizon physical residence fraction. The alternative C_T/T_*, for $T_* > 0$, describes the modeled interval and answers a different question.

For each species, define mass-density exposure

$$J_{j,T} = \int_0^{T_*} \rho_j(\mathbf{r}_t, t)\, dt. \tag{42}$$

Its dimensions are mass times inverse volume times time. A normalized exposure additionally requires a declared positive reference:

$$I_{j,T} = \int_0^{T_*} \frac{\rho_j(\mathbf{r}_t, t)}{\rho_{\text{ref},j}(t)}\, dt, \qquad \rho_{\text{ref},j}(t) > 0. \tag{43}$$

It has units of time. The reference is a convention, not an automatically inferred central density. Unless it bounds the sampled density, $I_{j,T}/T$ need not lie between zero and one.

Density may vary even while a trajectory remains inside $C(t)$; density outside that region may also be nonzero. Region-restricted exposure can be defined by inserting the same indicator into Eq. (42), but it is a distinct observable. None of these integrals is an encounter count.

C. Time-augmented occupation

Preserve the chronological relation between motion and environment through

$$\mu_T(\mathcal{A}) = \int_0^{T_*} \mathbf{1}_{\{(t,\mathbf{r}_t,\mathbf{v}_t)\in\mathcal{A}\}}\, dt, \tag{44}$$

where $\mathcal{A}$ is a measurable subset of $[0,T] \times \mathbb{R}^3 \times \mathbb{R}^3$. The measure has total mass T_* and units of time. Equations (41) and (42) are weighted integrals of that same measure:

$$C_T = \int \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, \mu_T(dt, d^3r, d^3v), \tag{45}$$

$$J_{j,T} = \int \rho_j(\mathbf{r}, t)\, \mu_T(dt, d^3r, d^3v). \tag{46}$$

Removing time labels can destroy the association with an evolving host; removing velocity can destroy information needed for a later dynamical kernel. Aggregated exposure therefore accompanies the trajectory and event record rather than replacing them.

D. Ensemble averages and incomplete histories

At fixed Q, nonnegative path integrands give the identities

$$\mathbb{E}[C_T \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, p_{\text{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q), \tag{47}$$

$$\mathbb{E}[J_{j,T} \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \rho_j(\mathbf{r}, t)\, p_{\text{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q). \tag{48}$$

If a density conditional on survival is used instead, it must be multiplied by $S(t \mid Q)$. Renormalizing the remaining trajectories silently changes the population being averaged. Equations (47) and (48) provide a future consistency comparison

between trajectory accumulation and a transport-density solution.

The event record must distinguish no exit observed, entry or re-entry observed, finite-horizon censoring, declared escape, unresolved strong interactions, and loss of model validity. Escape can compete with return; it is not interchangeable with numerical failure. A spatial boundary crossing counts as escape only under the declared escape rule.

After an unresolved stop, complete-horizon residence and exposure remain unknown. The nonnegative accumulated integrals describe the modeled interval, while extrapolation needs a continuation model. Censoring can depend on the path and environment. Treating it as independent would be an additional statistical assumption requiring justification.

VI. CONDITIONING AND CALCULATION REQUIREMENTS

A. Physical and inferential uncertainty

Host uncertainty, launch uncertainty, and physical scattering are different sources of variation. For a trajectory observable O, their joint accounting takes the form

$$\begin{aligned}\mathbb{E}[O \mid \mathcal{D}, \mathcal{M}] = &\int d\Theta\, p(\Theta \mid \mathcal{D}, \mathcal{M}) \\ &\times \int dz_0\, \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}) \\ &\times \mathbb{E}_{\text{path}}[O \mid \Theta, z_0, \mathcal{D}, \mathcal{M}]. \end{aligned} \tag{49}$$

The inner expectation includes the Wiener forcing and any additional randomness in a supplied stopping-event model. This factorization preserves conditional dependence. Merger selection can change host weights as well as launch properties. Alternative host families require separate conditioning or explicit model probabilities. Region definitions, reference densities, and any selection-dependent horizon must also be specified consistently.

Observed stellar kinematics constrain a host through an inference problem; they do not directly provide the entire potential, hidden remnant populations, or collision coefficients. The posterior summaries in Sec. III quantify dependence within published candidate fits. They do not determine probabilities for competing host families or their merger-conditioned weights.

For a fixed observed cluster c, a compatible reconstructed host law $\Pi_c(d\Theta \mid \mathcal{D}_c)$ would enter a trajectory-observable distribution as

$$\Pr_c(O \in A \mid \mathcal{D}_c, \lambda) = \int \Pr(O \in A \mid \Theta, \lambda)\, \Pi_c(d\Theta \mid \mathcal{D}_c). \tag{50}$$

Here A is a measurable set of observable values, not the drift vector $\mathbf{A}$, and λ retains the declared launch, recoil, collision, boundary, and horizon choices. The integrand includes the launch and path averaging in Eq. (49). This expression is a conditional modeling prescription, not an evaluated prediction. A merger-selected population additionally needs the host reweighting in Eq. (49); the six clusters are not assigned universal population weights.

Each host realization must supply the same F_j to Eqs. (9)–(13) and (25)–(28). The density used for exposure must come from that realization as well. At present, the projected data and candidate posterior records do not establish those local functions. Reproducing a source fit and its projected observables precedes any test of its suitability for drift and diffusion. Present-day observations also do not identify the stellar environment at a historical merger epoch.

B. Information passed between the two steps

A calculation becomes defined only when its physical inputs and event conventions are supplied. These include the units and frame; species masses and distributions; internal and external gravity with boundary conditions; the joint launch law; collision closure and scale cutoffs; physical horizon; region geometry; and escape, strong-event, and validity criteria. An unspecified input is not a zero-valued effect.

Step 1 must preserve physical times, positions, velocities, remnant-mass labels, host and launch identifiers, original realization weights, and termination causes. A density-based calculation must preserve its conditioning and absorbed or killed probability. Reproducibility requires the source of each input and enough numerical information to reconstruct coefficient evaluation, stochastic realizations, and error control.

Step 2 attaches crossing events, residence, species exposures, and a time-augmented occupation measure. Every summary retains its denominator convention, modeled endpoint, and status. A path that becomes unresolved is not filled by an assumed central rest state. The same host and epoch must supply the density used for exposure and the distribution used to drive transport.

C. Acceptance criteria before physical interpretation

The following requirements specify validation to be performed; they are not reported as executed checks. Dimensional consistency, force–potential agreement, number-to-mass normalization, and positive-semidefinite diffusion must first be established. The implementation must preserve the Itô convention and account for probability lost at each boundary or stop. Appropriate conservative and analytically justified stochastic limits provide independent checks, including the energy identity in Eq. (32).

Entry and residence calculations must distinguish tangencies from crossings, handle interior, exterior, and boundary launches, and account for moving regions. Brief passages must remain resolved under a declared refinement procedure. Convergence is needed for the relevant distributions and accumulated observables, not solely for a few representative trajectories. Numerical and sampling errors must be separated from uncertainty in physical inputs.

Where both representations exist, path averages and the survival-weighted identities in Eqs. (47) and (48) should agree

within declared numerical accuracy. Comparisons testing a physical simplification must hold the conditional population, horizon, region, and remaining inputs fixed. A useful result could identify a regime in which a reduction preserves the observables, or a discrepancy establishing which additional information is required.

Internal agreement does not calibrate the collision model. Comparison with independently resolved dynamics appropriate to the host and encounter regime is needed before interpreting predictions as quantitatively reliable. No solver, numerical tolerance, ensemble size, or passing threshold is assigned by the present formulation.

VII. DISCUSSION

The central object of the formulation is a merger-conditioned, time-resolved distribution of remnant motion. Finite-region entry, residence, and exposure are functionals of that distribution. Keeping their definitions separate makes explicit why remaining bound, returning to a region, and spending time in a dense environment are different physical questions.

The description permits off-center launches, inherited progenitor motion, anisotropic and streaming stellar populations, time-dependent gravity, and stochastic velocity changes. These dependences are available in the equations; they are not all measured or independently adjustable parameters. A physical realization must justify how they are constrained together and where the adopted closure applies.

The observational analysis makes several requirements concrete. Common units allow direct display comparisons, but source overlap and differing stellar selections prevent automatic construction of a shared likelihood. Strong mass–radius dependence in some published posteriors requires complete host realizations rather than independent marginal draws. Differences between structural estimates make the host prescription itself an uncertainty to examine. None of these findings alone supplies a local scattering distribution.

For a controlled comparison, define

$$\widetilde{C} = \frac{C_T}{t_0}, \qquad \widetilde{J_j} = \frac{J_{j,T} r_h^3}{M t_0}. \tag{51}$$

A testable next hypothesis is that host structure and stellar-mass composition leave differences in these dimensionless observables after fixing the dimensionless launch law, remnant-to-host mass ratio, T/t_0, region scaled by r_h, external-field treatment, collision prescription and termination conventions. The current analysis motivates this hypothesis by identifying structural variation and correlated host uncertainty; it does not test the transport consequence. A controlled comparison must first reproduce compatible host models, then compare the same dimensionless launch and event definitions while propagating each host's joint uncertainty. The gravitational scales alone cannot decide whether the resulting trajectory observables coincide.

The formulation supplies the conditional evolution and accumulation problem needed for that comparison. Continued motion after entry, survival weighting, and termination status remain explicit. Its connection to observed clusters is now supported by measured coverage and published host distributions, while calibrated friction, return, and residence predictions remain open calculations.

Timed position–velocity histories can eventually be combined with a separately justified companion population and interaction law. Residence and density exposure alone do not supply either ingredient. Companion acquisition, binary evolution, subsequent mergers, and inference from observed gravitational-wave sources therefore require additional physical descriptions beyond the two steps defined here.

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